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Nomiyama et al.

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(54) **LOW DROPOUT REGULATOR, CLOCK GENERATING CIRCUIT, AND MEMORY DEVICE**

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Dec. 27, 2022 (KR) 10-2022-0186388

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G11C 11/4074 (2006.01)
G11C 11/4076 (2006.01)

(52) **U.S. Cl.**
CPC ***G05F 1/575*** (2013.01); ***G11C 11/4074*** (2013.01); ***G11C 11/4076*** (2013.01)

(58) **Field of Classification Search**
CPC .. G05F 1/575; G11C 11/4074; G11C 11/4076
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

6,507,519	B1	1/2003	Collins et al.	
6,703,885	B1	3/2004	Fan et al.	
7,449,872	B2	11/2008	Zolfaghari	
9,122,293	B2	9/2015	Price et al.	
9,411,348	B2	8/2016	Iacob et al.	
10,310,528	B1	6/2019	Elsayed et al.	
11,003,201	B1 *	5/2021	Ahmed	G05F 1/59
2006/0273846	A1 *	12/2006	Lee	G11C 29/021 327/538
2009/0284246	A1 *	11/2009	Dash	G01R 31/40 324/76.11

FOREIGN PATENT DOCUMENTS

KR 10-2011-0113917 A 10/2011

* cited by examiner

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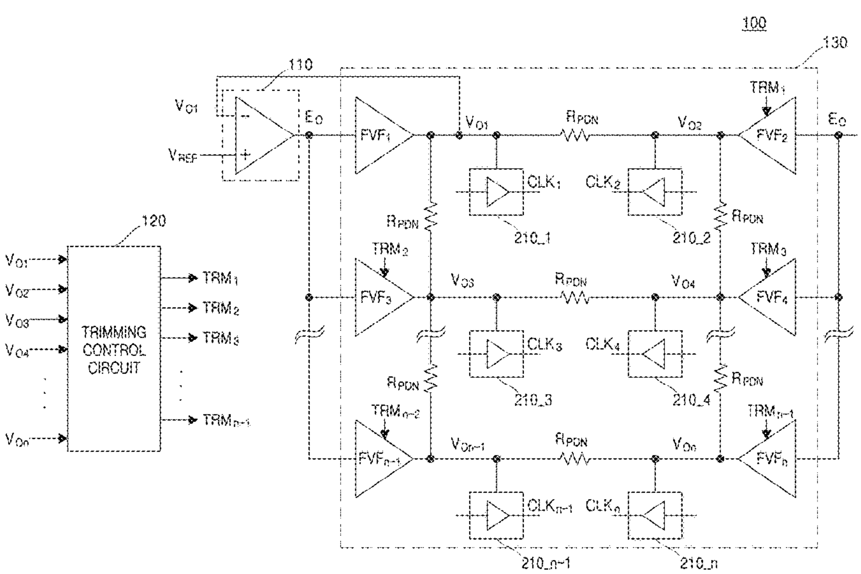
(57) **ABSTRACT**

A low dropout (LDO) regulator is configured to generate first to nth output voltages, where n is a natural number greater than or equal to 2, and each of the first to nth output voltages corresponds to a reference voltage. The LDO regulator includes an amplifier configured to generate an error voltage based on the reference voltage and a first output voltage of the first to nth output voltages, a trimming control circuit configured to generate first to (n−1)th trimming signals based on the first to nth output voltages, and an output buffer circuit configured to generate the first to nth output voltages based on the error voltage and the first to (n−1)th trimming signals.

20 Claims, 9 Drawing Sheets



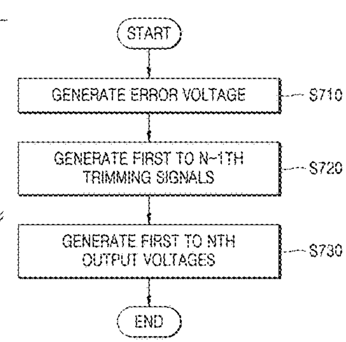


FIG. 1

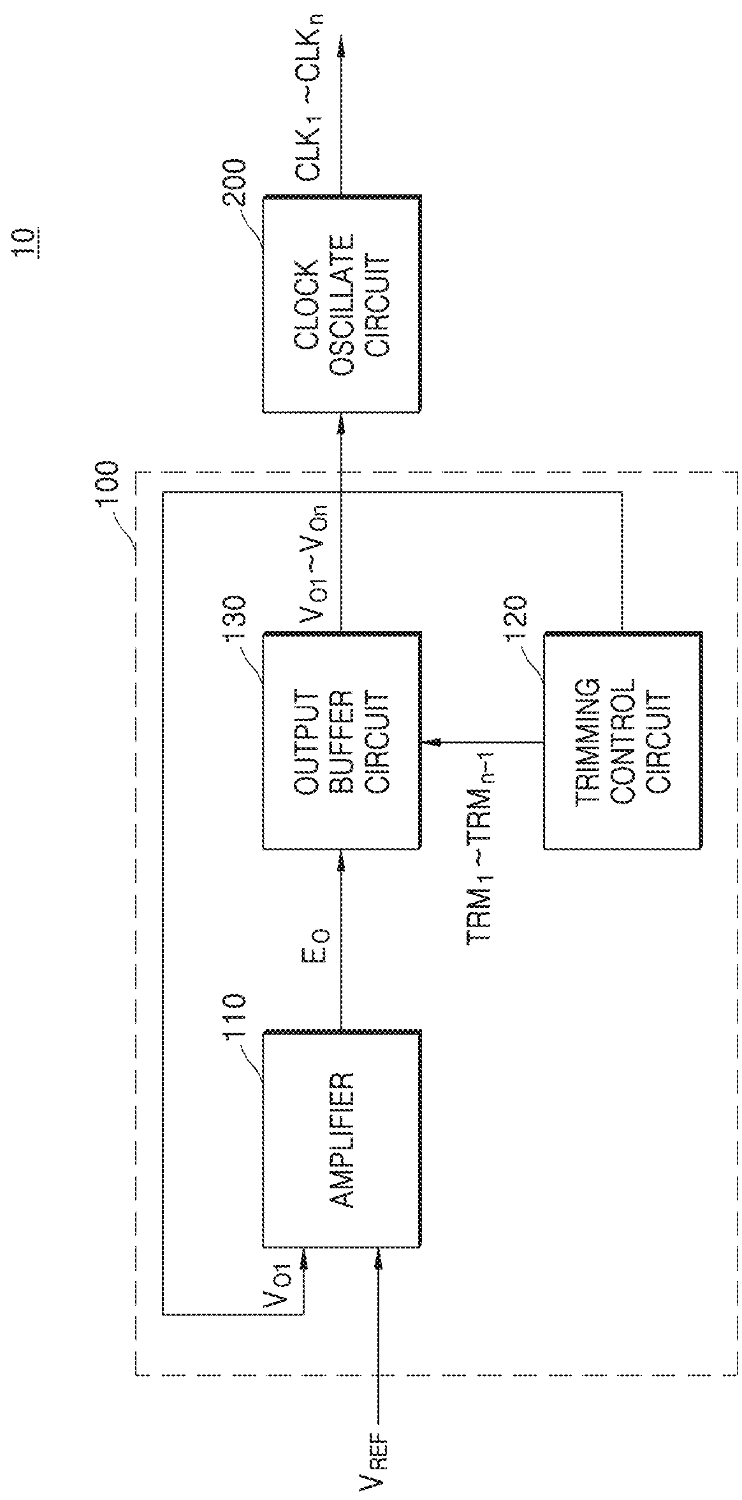

10
200
CLOCK OSCILLATE CIRCUIT
CLK_1 ~CLK_n
100
130
OUTPUT BUFFER CIRCUIT
V_{O1}~V_{On}
120
TRIMMING CONTROL CIRCUIT
TRM_1 ~TRM_{n-1}
110
AMPLIFIER
E_O
V_{O1}
V_{REF}

FIG. 2

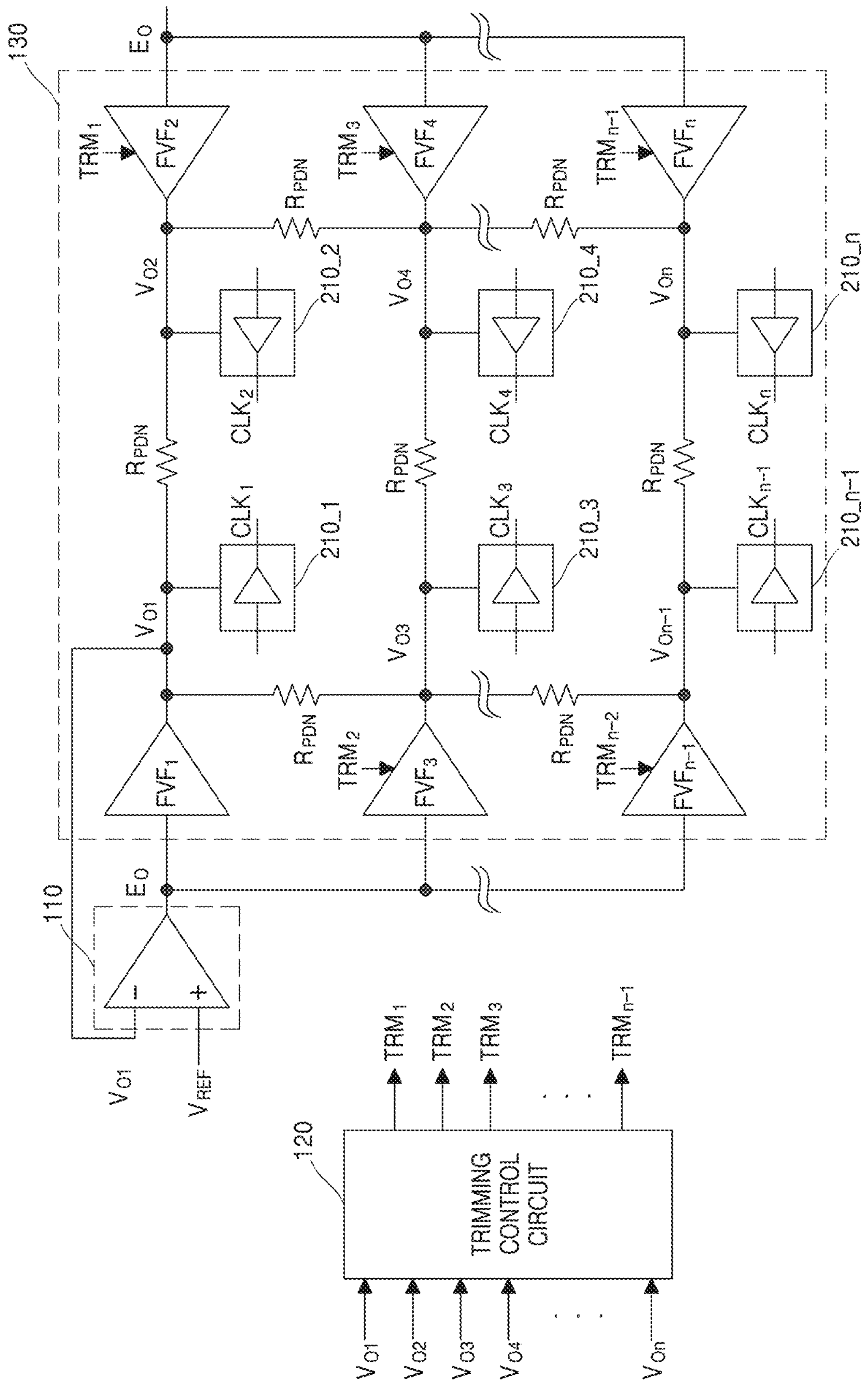

100
130
110
120
TRIMMING CONTROL CIRCUIT
E_O
V_{O1}
V_{REF}
V_{O1} V_{O2} V_{O3} V_{O4} V_{On}
TRM_1 TRM_2 TRM_3 TRM_{n-1}
FVF_1 FVF_2 FVF_3 FVF_4 FVF_{n-1} FVF_n
R_{PDN}
V_{O2} V_{O3} V_{O4} V_{On-1} V_{On}
CLK_1 CLK_2 CLK_3 CLK_4 CLK_{n-1} CLK_n
210_1 210_2 210_3 210_4 210_n−1 210_n

FIG. 3

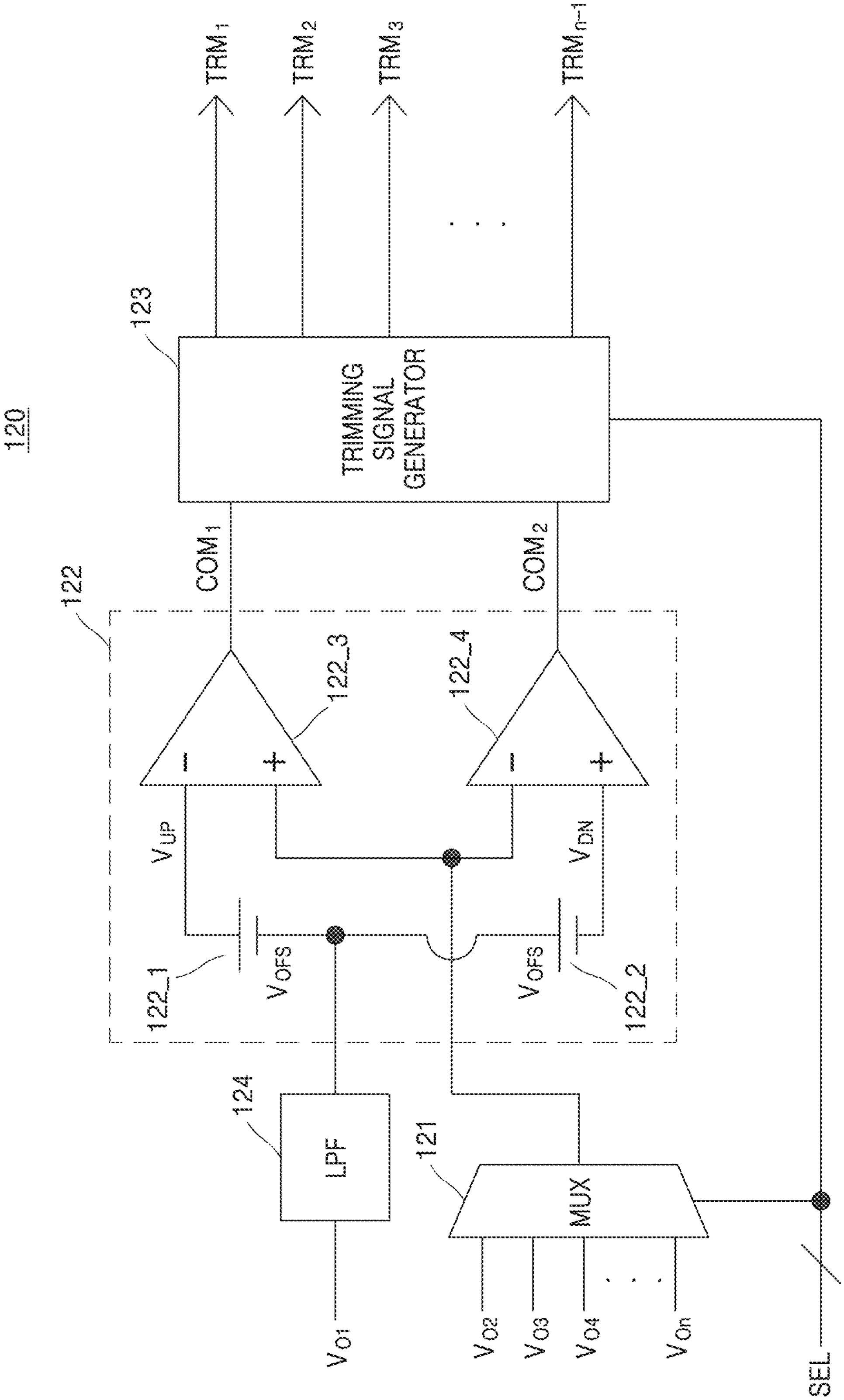

120
123
TRIMMING SIGNAL GENERATOR
TRM_1
TRM_2
TRM_3
TRM_{n-1}
122
COM_1
COM_2
122_3
122_4
V_{UP}
V_{DN}
122_1
V_{OFS}
V_{OFS}
122_2
124
LPF
121
MUX
V_{O1}
V_{O2}
V_{O3}
V_{O4}
V_{On}
SEL

FIG. 4

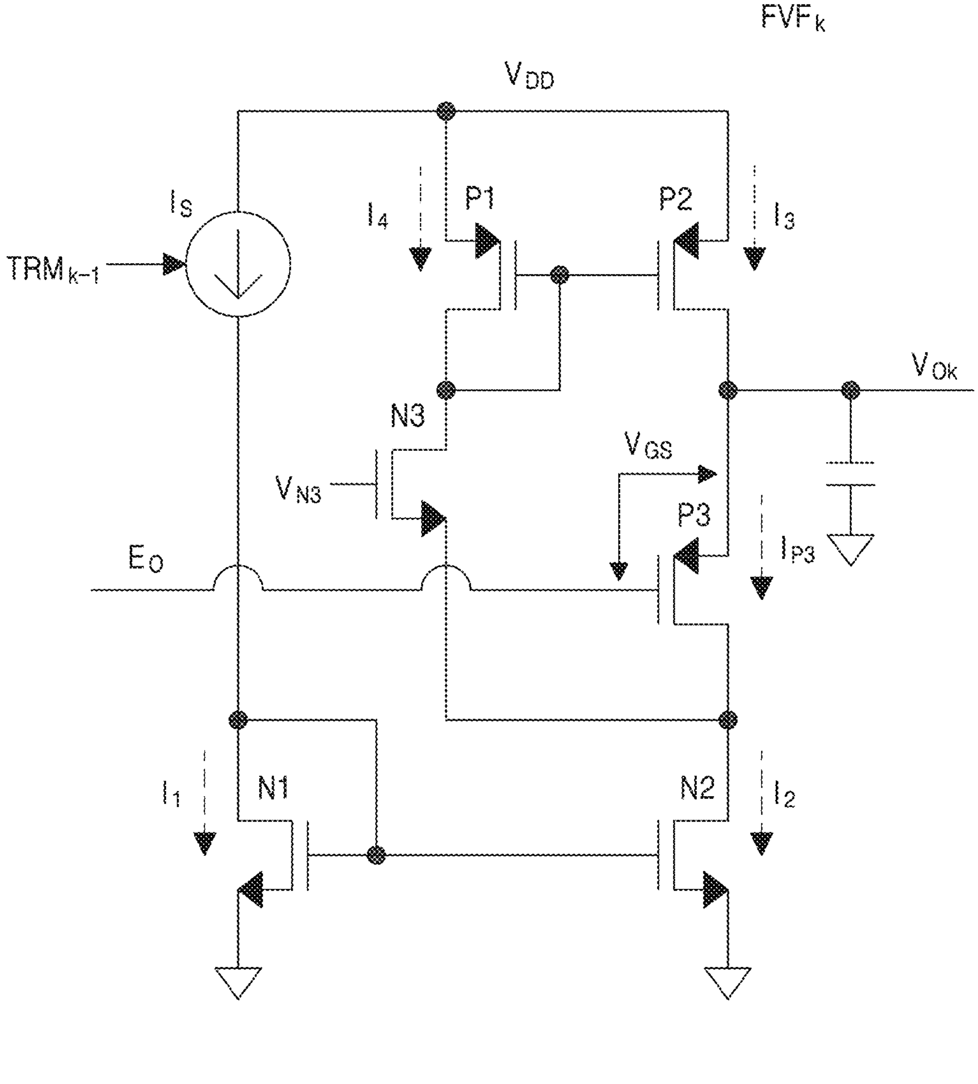


FIG. 5

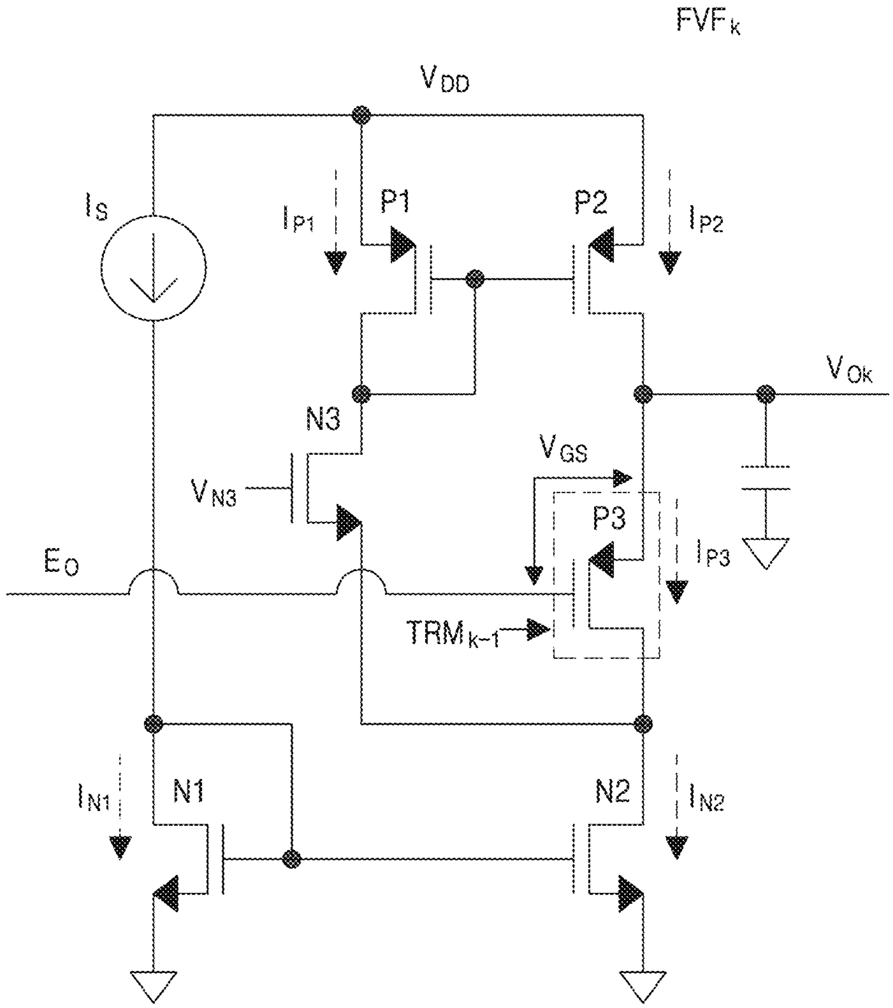
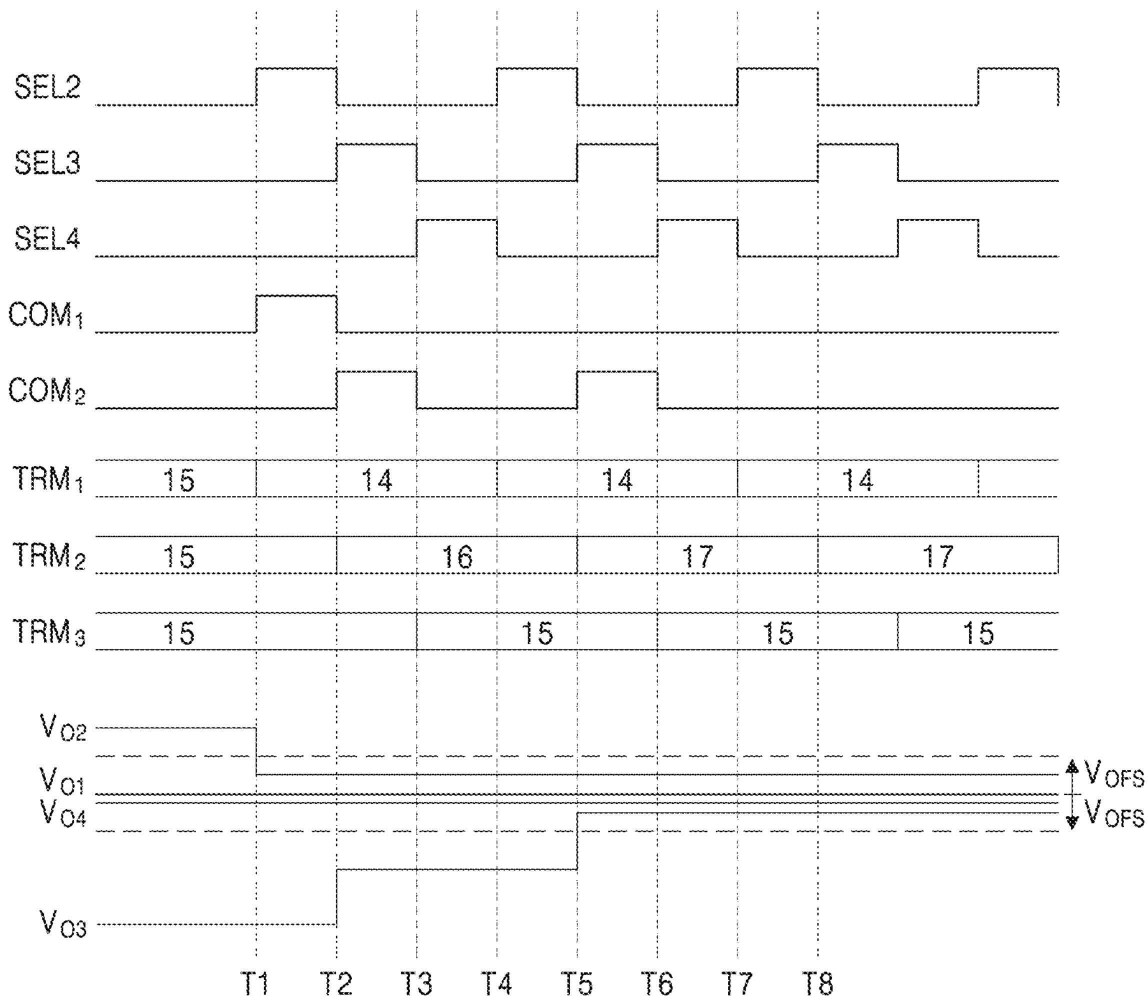

FVF_k
V_DD
I_S
I_P1
P1
P2
I_P2
V_Ok
N3
V_GS
V_N3
P3
I_P3
E_O
TRM_k-1
I_N1
N1
N2
I_N2

FIG. 6



SEL2
SEL3
SEL4
COM1
COM2
TRM1
15
14
14
14
TRM2
15
16
17
17
TRM3
15
15
15
15
VO2
VO1
VO4
VO3
VOFS
VOFS
T1
T2
T3
T4
T5
T6
T7
T8

FIG. 7

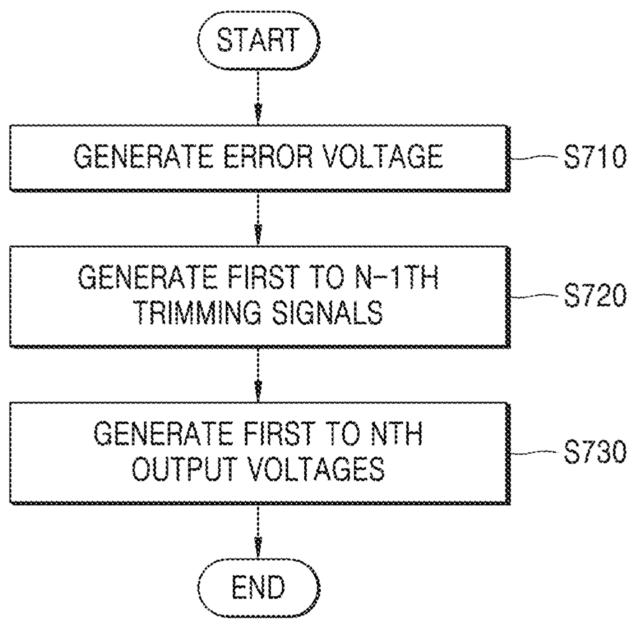

START
GENERATE ERROR VOLTAGE
S710
GENERATE FIRST TO N-1TH TRIMMING SIGNALS
S720
GENERATE FIRST TO NTH OUTPUT VOLTAGES
S730
END

FIG. 8

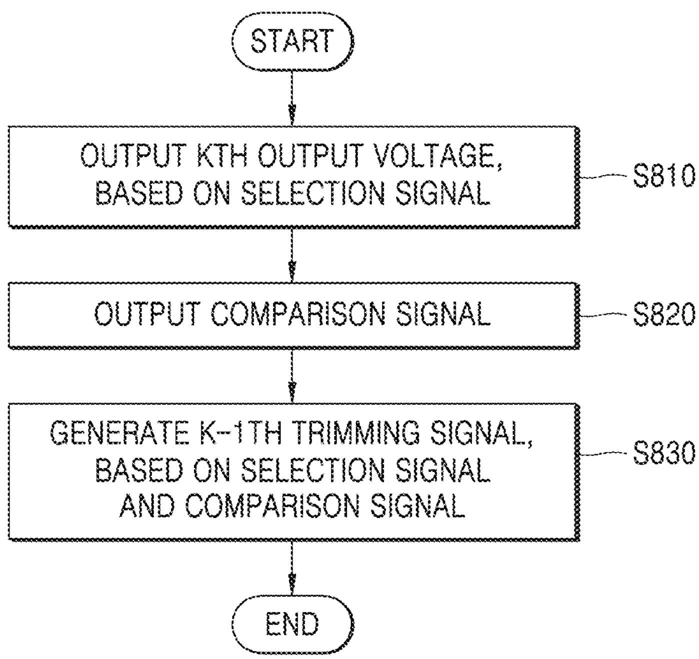
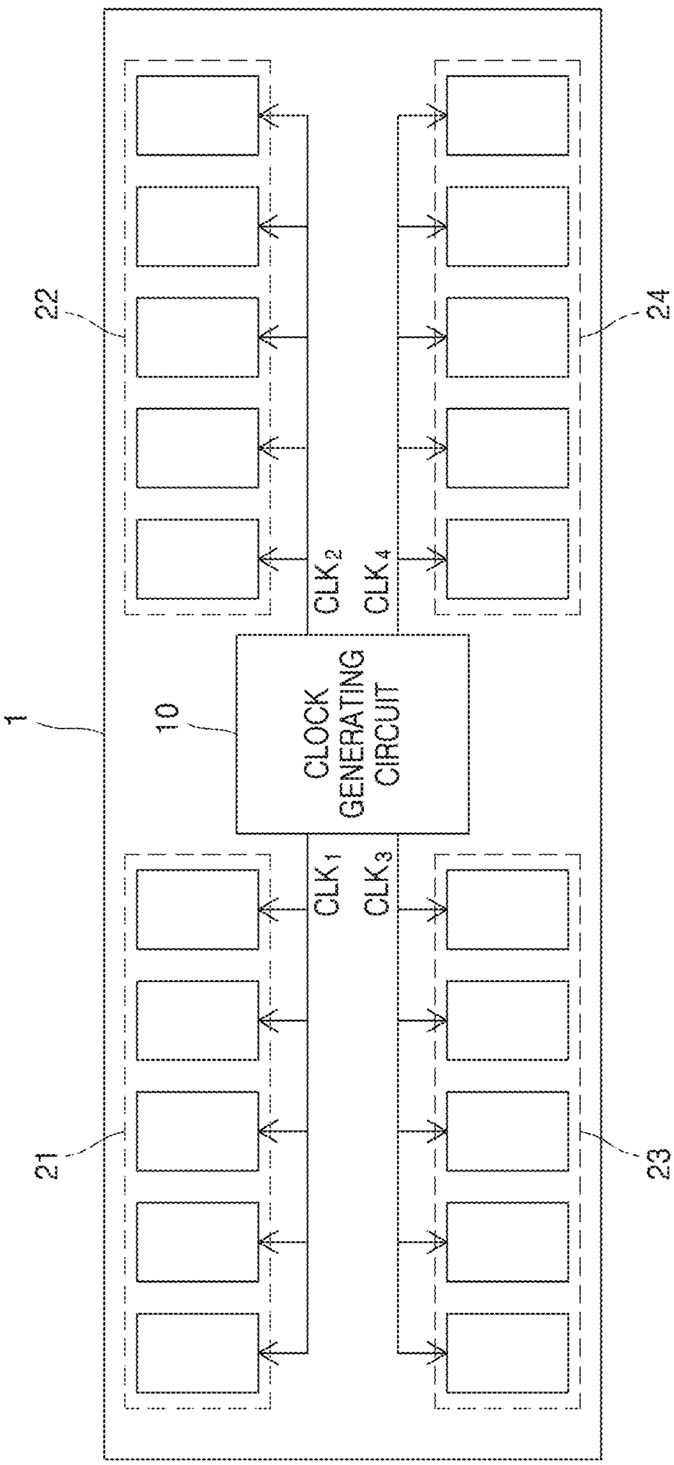

START
OUTPUT KTH OUTPUT VOLTAGE, BASED ON SELECTION SIGNAL
S810
OUTPUT COMPARISON SIGNAL
S820
GENERATE K-1TH TRIMMING SIGNAL, BASED ON SELECTION SIGNAL AND COMPARISON SIGNAL
S830
END

FIG. 9



LOW DROPOUT REGULATOR, CLOCK GENERATING CIRCUIT, AND MEMORY DEVICE

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2022-0137764, filed on Oct. 24, 2022, and Korean Patent Application No. 10-2022-0186388, filed on Dec. 27, 2022, in the Korean Intellectual Property Office, the disclosures of each of which are incorporated by reference herein in their entireties.

FIELD

The inventive concepts relate to a low dropout (LDO) regulator, including an LDO regulator configured to generate an output voltage based on a trimming signal.

BACKGROUND

As technology advances, a number of chips for performing various kinds of operations are equipped in one electronic device. In this case, in order to operate the chips included in an electronic device at an accurate timing, an input of an accurate clock signal may be desired or needed. For example, modules such as dual in-line memory modules (DIMMs) each include a plurality of dynamic random access memory (DRAM) chips, and an input of an accurate clock signal may be desired or needed for accurate operation of each DRAM chip.

Clock generating circuits may generate a clock signal based on a voltage applied thereto. In this case, power conversion circuits such as LDO regulators may be used for adjusting a level of a voltage applied to a clock generating circuit. In this case, LDO regulators may apply a plurality of output voltages to a clock generating circuit through an output buffer. However, a voltage applied to a clock generating circuit may be changed due to causes such as current consumption and an internal resistance caused by a line included in an output buffer. When a voltage applied to a clock generating circuit is changed, clock skew between a plurality of clock signals generated through a clock generating circuit may occur. Clock skew may be a cause of malfunction of an electronic device such as a communication error. Therefore, the development of a method for solving such a problem may be desired or needed.

SUMMARY

The inventive concepts provide a low dropout (LDO) regulator which may apply a certain voltage to a clock generating circuit.

According to an aspect of the inventive concepts, a low dropout (LDO) regulator is configured to generate first to nth output voltages, where n is a natural number greater than or equal to 2, and each of the first to nth output voltages corresponds to a reference voltage. The LDO regulator includes an amplifier configured to generate an error voltage based on the reference voltage and a first output voltage of the first to nth output voltages, a trimming control circuit configured to generate first to (n−1)th trimming signals based on the first to nth output voltages, and an output buffer circuit configured to generate the first to nth output voltages based on the error voltage and the first to (n−1)th trimming signals.

According to another aspect of the inventive concepts, a clock generating circuit includes a low dropout (LDO) regulator configured to generate a plurality of output voltages each corresponding to a reference voltage, and a clock oscillate circuit configured to generate a plurality of clock signals based on the plurality of output voltages. The LDO regulator includes an amplifier configured to generate an error voltage based on the reference voltage and a first output voltage of the plurality of output voltages, a trimming control circuit configured to generate a plurality of trimming signals based on the plurality of output voltages, and an output buffer circuit configured to generate the plurality of output voltages based on the error voltage and the plurality of trimming signals.

According to another aspect of the inventive concepts, a memory device includes a clock generating circuit configured to generate a plurality of clock signals, and a plurality of dynamic random access memory (DRAM) chips configured to operate based on the plurality of clock signals. The clock generating circuit includes a low dropout (LDO) regulator configured to generate a plurality of output voltages each corresponding to a reference voltage, and a clock oscillate circuit configured to generate a plurality of clock signals, based on the plurality of output voltages. The LDO regulator includes an amplifier configured to generate an error voltage based on the reference voltage and a first output voltage of the plurality of output voltages, a trimming control circuit configured to generate a plurality of trimming signals based on the plurality of output voltages, and an output buffer circuit configured to generate the plurality of output voltages based on the error voltage and the plurality of trimming signals.

BRIEF DESCRIPTION OF THE DRAWINGS

Example embodiments will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings in which:

FIG. **1** is a block diagram illustrating a clock generating circuit according to an example embodiment;

FIG. **2** is a circuit diagram illustrating a low dropout (LDO) regulator according to an example embodiment;

FIG. **3** is a circuit diagram illustrating a detailed configuration of a trimming control circuit of an LDO regulator according to an example embodiment;

FIG. **4** is a circuit diagram illustrating an example embodiment of a flipped voltage follower (FVF) buffer of an LDO regulator;

FIG. **5** is a circuit diagram illustrating another example embodiment of an FVF buffer of an LDO regulator;

FIG. **6** is a timing diagram illustrating a voltage and a signal of a trimming control circuit of an LDO regulator according to an example embodiment;

FIG. **7** is a flowchart illustrating an operating method of an LDO regulator, according to an example embodiment;

FIG. **8** is a flowchart illustrating an operating method of a trimming control circuit of an LDO regulator, according to an example embodiment; and

FIG. **9** is a block diagram illustrating a memory device according to an example embodiment.

DETAILED DESCRIPTION

Hereinafter, example embodiments will be described in detail with reference to the accompanying drawings.

FIG. **1** is a block diagram illustrating a clock generating circuit **10** according to an example embodiment.

Referring to FIG. **1**, the clock generating circuit **10** according to an example embodiment may include a low dropout (LDO) regulator **100** and a clock oscillate circuit **200**.

The clock generating circuit **10** may generate a plurality of clock signals CLK_1 to CLK_n. The number of clock signals CLK_1 to CLK_n may be an n (where n is a natural number of 2 or more) number, and the plurality of clock signals CLK_1 to CLK_n may include first to nth clock signals CLK_1 to CLK_n. The clock generating circuit **10** may output the generated plurality of clock signals CLK_1 to CLK_n to other devices (for example, dynamic random access memory (DRAM) chips). In some example embodiments, the plurality of clock signals CLK_1 to CLK_n generated by the clock generating circuit **10** may be used as a signal which is a criterion of an operation in the other devices. Therefore, it may be desirable for the clock generating circuit to generate the plurality of clock signals CLK_1 to CLK_n without an error such as clock skew.

The LDO regulator **100** may generate a plurality of output voltages V_{O1} to V_{On} corresponding to a reference voltage V_{REF}. The number of output voltages V_{O1} to V_{On} may be an n number which is equal to the number of clock signals CLK_1 to CLK_n, and the plurality of output voltages V_{O1} to V_{On} may include first to nth output voltages V_{O1} to V_{On}. For example, the LDO regulator **100** may generate the plurality of output voltages V_{O1} to V_{On} having the same or substantially the same level as that of the reference voltage V_{REF}.

The LDO regulator **100** may include an amplifier **110**, a trimming control circuit **120**, and an output buffer circuit **130**.

The amplifier **110** may generate an error voltage E_O, based on the reference voltage V_{REF} and the first output voltage V_{O1}. The amplifier **110** may amplify a difference between the reference voltage V_{REF} and the first output voltage V_{O1} to generate the error voltage E_O. For example, the amplifier **110** may subtract the reference voltage V_{REF} from the first output voltage V_{O1} and may amplify a subtraction result by a desired (or alternatively, predetermined) rate to generate the error voltage E_O.

The amplifier **110** may receive the first output voltage V_{O1} from the output buffer circuit **130** described below. In some example embodiments, the first output voltage V_{O1} may be a voltage which is generated by the output buffer circuit **130** at a previous stage.

The reference voltage V_{REF} may be a voltage corresponding to a voltage which is to be generated by the LDO regulator **100**. In an example embodiment, the reference voltage V_{REF} may be a voltage which is received from an external host device. In another example embodiment, the reference voltage V_{REF} may be a voltage which is stored in the LDO regulator **100**.

The trimming control circuit **120** may generate a plurality of trimming signals TRM_1 to TRM_{n-1}, based on the plurality of output voltages V_{O1} to V_{On}.

The trimming control circuit **120** may receive the plurality of output voltages V_{O1} to V_{On} from the output buffer circuit **130** described below. In some example embodiments, the plurality of output voltages V_{O1} to V_{On} may each be a voltage which is generated by the output buffer circuit **130** at a previous stage.

The plurality of trimming signals TRM_1 to TRM_{n-1} may be used to generate the plurality of output voltages V_{O1} to V_{On} in the output buffer circuit **130** described below. In an example embodiment, the plurality of trimming signals TRM_1 to TRM_{n-1} may each be a signal for control which is performed so that levels of the plurality of output voltages V_{O1} to V_{On} are maintained to be equal or substantially equal to one another in the output buffer circuit **130**.

The number of may be an n−1 number which is one less than the number of output voltages V_{O1} to V_{On} and may include first to n−1th trimming signals TRM_1 to TRM_{n-1}. In some example embodiments, the first trimming signal TRM_1 may be a signal for control which is performed so that levels of the first output voltage V_{O1} and the second output voltage V_{O2} are maintained to be equal to each other. Also, the k−1th trimming signal TRM_{k-1} (where k is a natural number of 2 or more and n or less) may be a signal for control which is performed so that levels of the first output voltage V_{O1} and the kth output voltage V_{Ok} are maintained to be equal or substantially equal to each other. Also, the n−1th trimming signal TRM_{n-1} may be a signal for control which is performed so that levels of the first output voltage V_{O1} and the nth output voltage V_{On} are maintained to be equal to each other.

A more detailed structure and operation of the trimming control circuit **120** will be described below with reference to FIG. **3**.

The output buffer circuit **130** may generate the plurality of output voltages V_{O1} to V_{On}, based on the error voltage E_O and the plurality of trimming signals TRM_1 to TRM_{n-1}. A more detailed structure and operation of the output buffer circuit **130** will be described below with reference to FIG. **2**.

The clock oscillate circuit **200** may generate a plurality of clock signals CLK_1 to CLK_n, based on the plurality of output voltages V_{O1} to V_{On}. The clock oscillate circuit **200** may include a plurality of oscillators and may respectively generate the plurality of clock signals CLK_1 to CLK_n by using the plurality of oscillators, based on the plurality of output voltages V_{O1} to V_{On}. In some example embodiments, the number of oscillators may be equal to the number of output voltages V_{O1} to V_{On}, and a kth oscillator may generate a kth clock signal CLK_k, based on the kth output voltage V_{Ok}.

FIG. **2** is a circuit diagram illustrating an LDO regulator **100** according to an example embodiment.

Referring to FIG. **2**, the LDO regulator **100** according to an example embodiment may include an amplifier **110**, a trimming control circuit **120**, and an output buffer circuit **130**.

The amplifier **110** may generate an error voltage E_O, based on a reference voltage V_{REF} and a first output voltage V_{O1}. In an example embodiment, the amplifier **110** may be an error amplifier.

The trimming control circuit **120** may generate a plurality of trimming signals TRM_1 to TRM_{n-1}, based on a plurality of output voltages V_{O1} to V_{On}. A more detailed structure and operation of the trimming control circuit **120** will be described below with reference to FIG. **3**.

The output buffer circuit **130** may generate the plurality of output voltages V_{O1} to V_{On}, based on the error voltage E_O and the plurality of trimming signals TRW to TRM_{n-1}.

The output buffer circuit **130** may include a plurality of flipped voltage follower (FVF) buffers FVF_1 to FVF_n.

In FIG. **2**, the output buffer circuit **130** is illustrated as including a plurality of power delivery network (PDN) resistors R_{PDN}, but each of the plurality of PDN resistors R_{PDN} may not be a resistor actually connected between output buffer circuits **130**, and may be an internal resistor caused by a line included in the output buffer circuit **130**. Due to the plurality of PDN resistors R_{PDN}, when the plurality of FVF buffers FVF_1 to FVF_n are implemented to be equal or substantially equal to one another and the same error voltage E_O is received, voltages having different levels may be output.

Also, in FIG. **2**, a plurality of oscillators **210_1** to **210_n** included in the output buffer circuit **130** are illustrated, and in some example embodiments the plurality of oscillators **210_1** to **210_n** may be elements included in the clock oscillate circuit **200** of FIG. **1**.

The plurality of FVF buffers FVF_1 to FVF_n may generate the plurality of output voltages V_{O1} to V_{On}, based on the error voltage E_O and the plurality of trimming signals TRM_1 to TRM_{n-1}. The number of FVF buffers FVF_1 to FVF_n may be an n number which is equal to the number of output voltages V_{O1} to V_{On}, and may include first to nth FVF buffers FVF_1 to FVF_n.

The plurality of FVF buffers FVF_1 to FVF_n may receive the error voltage E_O output from the amplifier **110**. At this time, the first FVF buffer FVF_1 may not receive the plurality of trimming signals TRM_1 to TRM_{n-1}. This may be because the plurality of output voltages V_{O1} to V_{On} are controlled to be equal or substantially equal to a level of the first output voltage V_{O1}. Also, the second to nth FVF buffers FVF_2 to FVF_n may receive the first to n−1th trimming signals TRM_1 to TRM_{n-1}. For example, the kth FVF buffer FVF_k may receive the k−1th trimming signal TRM_{k-1}.

The first FVF buffer FVF_1 may generate the first output voltage V_{O1}, based on the error voltage E_O.

The second to nth FVF buffers FVF_2 to FVF_n may generate the second to nth output voltages V_{O2} to V_{On}, based on the error voltage E_O and the first to n−1th trimming signals TRM_1 to TRM_{n-1}. For example, the kth FVF buffer FVF_k may generate the kth output voltage V_{Ok}, based on the error voltage E_O and the k−1th trimming signal TRM_{k-1}.

In an example embodiment, the second to nth FVF buffers FVF_2 to FVF_n may respectively control a bias current, based on the first to n−1th trimming signals TRM_1 to TRM_{n-1}, and may generate the second to nth output voltages V_{O2} to V_{On}, based on the controlled bias current. An example embodiment where the second to nth FVF buffers FVF_2 to FVF_n control the bias current will be described below in more detail with reference to FIG. **4**.

In an example embodiment, the second to nth FVF buffers FVF_2 to FVF_n may respectively control connections between a plurality of target switching elements, based on the first to n−1th trimming signals TRM_1 to TRM_{n-1}, and may generate the second to nth output voltages V_{O2} to V_{On}, based on voltages by the plurality of target switching elements. An example embodiment where the second to nth FVF buffers FVF_2 to FVF_n control the connections between the plurality of target switching elements will be described below in more detail with reference to FIG. **5**.

FIG. **3** is a circuit diagram illustrating a detailed configuration of a trimming control circuit **120** of an LDO regulator according to an example embodiment.

Referring to FIG. **3**, the trimming control circuit **120** according to an embodiment may include a multiplexer (MUX) **121**, a comparison circuit **122**, and a trimming signal generator **123**. According to an example embodiment, the trimming control circuit **120** may further include a low pass filter (LPF) **124**.

The multiplexer **121** may output one of second to nth output voltages V_{O2} to V_{On}, based on a selection signal SEL.

The selection signal SEL may be a signal which is used for driving of the multiplexer **121** in the trimming control circuit **120**. In an example embodiment, the selection signal SEL may be previously set so that multiplexer **121** alternately selects the second to nth output voltages V_{O2} to V_{On}.

The multiplexer **121** may sequentially output the second to nth output voltages V_{O2} to V_{On}, based on the selection signal SEL. That is, the multiplexer **121** may sequentially output the second output voltage V_{O2}, and the third output voltage V_{O3}, and the fourth output voltage V_{O4}, based on the selection signal SEL.

The comparison circuit **122** may compare the first output voltage V_{O1} with the kth output voltage V_{Ok} output by the multiplexer **121** to output comparison signals COM_1 and COM_2. In an example embodiment, the comparison signals COM_1 and COM_2 may include a first comparison signal COM_1 and a second comparison signal COM_2. The comparison circuit **122** may output, as the comparison signals COM_1 and COM_2, a result obtained by determining whether the kth output voltage V_{Ok} output by the multiplexer **121** is within a desired (or alternatively, predetermined) reference error range with respect to the first output voltage V_{O1}.

The comparison circuit **122** may include a first power source **122_1**, a second power source **122_2**, a first comparator **122_3**, and a second comparator **122_4**.

The first power source **122_1** may summate the first output voltage V_{O1} and an offset voltage V_{OFS} to output an upper voltage V_{UP}.

In more detail, the first power source **122_1** may be a power element having a level of the offset voltage V_{OFS}. The offset voltage V_{OFS} may be set to a value corresponding to half of the reference error range. The first power source **122_1** may receive the first output voltage V_{O1} through a negative terminal. Also, the first power source **122_1** may output, through a positive terminal, the upper voltage V_{UP} having a level which is a sum of the first output voltage V_{O1} and the offset voltage V_{OFS}. At this time, a positive terminal of the first power source **122_1** may be connected with the first comparator **122_3** described below.

The second power source **122_2** may subtract the offset voltage V_{OFS} from the first output voltage V_{O1} to output a lower voltage V_{DN}.

In more detail, the second power source **122_2** may be a power element having a level of the offset voltage V_{OFS}. In some example embodiments, the second power source **122_2** may be a power element having a voltage having the same or substantially the same level as that of the first power source **122_1**. The second power source **122_2** may receive the first output voltage V_{O1} through a positive terminal. Also, the second power source **122_2** may output, through a negative terminal, the lower voltage V_{DN} having a level obtained by subtracting the offset voltage V_{OFS} from the first output voltage V_{O1}. At this time, a negative terminal of the second power source **122_2** may be connected with the second comparator **122_4** described below.

The first comparator **122_3** may compare the upper voltage V_{UP} with the kth output voltage V_{Ok} to generate the first comparison signal COM_1. The first comparison signal COM_1 may be a signal representing a result obtained by determining whether the kth output voltage V_{Ok} is greater than an upper limit of the reference error range. For example, the first comparator **122_3** may subtract the upper voltage V_{UP} from the kth output voltage V_{Ok} and may generate the first comparison signal COM_1, based on whether a subtraction result has a positive value.

When the kth output voltage V_{Ok} is greater than the upper voltage V_{UP}, the first comparator **122_3** may output the first comparison signal COM_1 having a first value (for example, logic 1). On the other hand, when the kth output voltage V_{Ok} is less than or equal to the upper voltage V_{UP}, the first comparator **122_3** may output the first comparison signal COM_1 having a second value (for example, logic 0).

The second comparator **122_4** may compare the lower voltage V_{DN} with the kth output voltage V_{Ok} to generate the second comparison signal COM_2. The second comparison signal COM_2 may be a signal representing a result obtained by determining whether the kth output voltage V_{Ok} is greater than a lower limit of the reference error range. For example, the second comparator **122_4** may subtract the lower voltage V_{DN} from the kth output voltage V_{Ok} and may generate the second comparison signal COM_2, based on whether a subtraction result has a positive value.

When the kth output voltage V_{Ok} is less than the lower voltage V_{DN}, the second comparator **122_4** may output the second comparison signal COM_2 having a third value (for example, logic 1). On the other hand, when the kth output voltage V_{Ok} is greater than or equal to the lower voltage V_{DN}, the second comparator **122_4** may output the second comparison signal COM_2 having a fourth value (for example, logic 0).

The trimming signal generator **123** may generate the k−1th trimming signal TRM_{k-1}, based on the selection signal SEL and the comparison signals COM_1 and COM_2.

In an example embodiment, when the first comparison signal COM_1 has the first value, the trimming signal generator **123** may generate the k−1th trimming signal TRM_{k-1} for control which is performed to reduce the kth output voltage V_{Ok}. Because a case where the first comparison signal COM_1 has the first value denotes that the kth output voltage V_{Ok} is greater than the upper limit of the reference error range, the trimming signal generator **123** may generate the k−1th trimming signal TRM_{k-1} for control which is performed to reduce the kth output voltage V_{Ok}.

In an example embodiment, when the second comparison signal COM_2 has the third value, the trimming signal generator **123** may generate the k−1th trimming signal TRM_{k-1} for control which is performed to increase the kth output voltage V_{Ok}. Because a case where the second comparison signal COM_2 has the third value denotes that the kth output voltage V_{O1} (is smaller than the lower limit of the reference error range, the trimming signal generator **123** may generate the k−1th trimming signal TRM_{k-1} for control which is performed to increase the kth output voltage V_{Ok}.

In an example embodiment, when where the first comparison signal COM_1 has the second value and the second comparison signal COM_2 has the fourth value, the trimming signal generator **123** may generate the k−1th trimming signal TRM_{k-1} for control which is performed to intactly maintain the kth output voltage V_{Ok}. Because a case where the first comparison signal COM_1 has the second value and the second comparison signal COM_2 has the fourth value denotes that the kth output voltage V_{O1} (is within the reference error range, the trimming signal generator **123** may generate the k−1th trimming signal TRM_{k-1} for control which is performed to intactly maintain the kth output voltage V_{Ok}.

The trimming signal generator **123** may receive the same selection signal SEL simultaneously with the multiplexer **121**. For example, when a selection signal for allowing the kth output voltage V_{O1} (to be output is input to the multiplexer **121**, the trimming signal generator **123** may receive the same selection signal SEL simultaneously or substantially simultaneously, and thus, may generate the k−1th trimming signal TRM_{k-1} for control which is performed so that a level of the first output voltage V_{O1} and a level of the kth output voltage V_{O1} (are maintained to be equal to each other. Therefore, the trimming signal generator **123** may sequentially generate the first to n−1th trimming signals TRM_1 to TRM_{n-1}, based on the selection signal SEL. That is, the trimming signal generator **123** may sequentially output the first trimming signal TRM_1, the second trimming signal TRM_2, and the third trimming signal TRM_3, based on the selection signal SEL.

The low pass filter **124** may remove noise included in the first output voltage V_{O1} and may input the noise-removed first output voltage V_{O1}. The low pass filter **124** may receive the first output voltage V_{O1}. The low pass filter **124** may remove high frequency noise included in the received first output voltage V_{O1}. The low pass filter **124** may output the noise-removed first output voltage V_{O1} to the comparison circuit **122**.

FIG. **4** is a circuit diagram illustrating an embodiment of an FVF buffer of an LDO regulator **100** according to an example embodiment.

Referring to FIG. **4**, an example of the circuit implementation of second to nth FVF buffers FVF_2 to FVF_n included in an output buffer circuit **130** included in the LDO regulator **100** according to an example embodiment may be seen.

In the example embodiment of FIG. **4**, the kth FVF buffer FVF_k may include a current source I_S, a first switching element **N1**, a second switching element **N2**, a third switching element **N3**, a fourth switching element **P1**, a fifth switching element **P2**, and a sixth switching element **P3**.

The current source I_S may supply a bias current to the first switching element **N1**. Therefore, a first current I_1 may be applied to one terminal of the first switching element **N1**. The first switching element **N1** and the second switching element **N2** may operate as a current mirror. Accordingly, a second current I_2 having a desired (or alternatively, predetermined) ratio to the first current I_1 may flow in the second switching element **N2**.

The third switching element **N3** may operate based on an input voltage V_{N3}.

The same third current I_3 as an output current may flow in the fifth switching element **P2**. At this time, the fourth switching element **P1** and the fifth switching element **P2** may operate as a current mirror. Accordingly, a fourth current I_4 having a desired (or alternatively, predetermined) ratio to the third current I_3 may flow in the fourth switching element **P1**.

An error voltage E_O may be input to a gate terminal of the sixth switching element **P3**. A voltage input to the gate terminal of the sixth switching element **P3** may increase by a gate-source voltage V_{GS}, and thus, the kth output voltage V_{Ok} may be generated.

In some example embodiments, the kth output voltage V_{Ok} may be expressed as in the following Equation 1.

$$V_{Ok} = E_O + V_{GS} = E_O + \sqrt{\frac{2\times\left(I_1\times\frac{B}{A} - I_3\times\frac{C}{D}\right)}{\beta_{P3}\times\frac{W_{P3}}{L_{P3}}}} + V_{TH_{P3}} \quad \text{[Equation 1]}$$

In Equation 1, B/A may denote a current ratio between the first switching element **N1** and the second switching element **N2**, and C/D may denote a current ratio between the fourth switching element **P1** and the fifth switching element **P2**. Also, β_{P3} may denote a current gain of the sixth switching element **P3**, W_{P3} may denote a width of the sixth switching element **P3**, L_{P3} may denote a length of the sixth switching element **P3**, and V_{THP3} may denote a threshold voltage of the sixth switching element **P3**.

In the example embodiment of FIG. **4**, the k−1th trimming signal TRM_{k-1} may be applied to the current source I_S. At this time, the current source I_S may adjust a bias current, based on the k−1th trimming signal TRM_{k-1}. The bias current of the current source I_S may be the first current I_1 and may be applied to one terminal of the first switching element **N1**.

When the k−1th trimming signal TRM_{k-1} is a signal for control performed to increase the kth output voltage V_{Ok}, the current source I_S may increase the bias current. Therefore, the first current I_1 may increase. In some example embodiments, as in Equation 1, the kth output voltage V_{Ok} may be proportional to the first current I_1, and thus, the kth output voltage V_{Ok} may increase.

When the k−1th trimming signal TRM_{k-1} is a signal for control performed to decrease the kth output voltage V_{Ok}, the current source I_S may decrease the bias current. Therefore, the first current I_1 may decrease. In some example embodiments, as in Equation 1, the kth output voltage V_{Ok} may be proportional to the first current I_1, and thus, the kth output voltage V_{Ok} may decrease.

When the k−1th trimming signal TRM_{k-1} is a signal for control performed to intactly maintain the kth output voltage V_{Ok}, the current source I_S may intactly maintain the bias current. Therefore, the first current I_1 may be intactly maintained, and the kth output voltage V_{Ok} may be intactly maintained.

As described above, the LDO regulator **100** according to an example embodiment may generate a plurality of trimming signals TRM_1 to TRM_{n-1} generated by the trimming control circuit **120**, based on a plurality of output voltages V_{O1} to V_{On}, and may adjust a bias current based on the current source I_S included in the second to nth FVF buffers FVF_2 to FVF_n of the output buffer circuit **130** to control constant levels of the second to nth FVF buffers FVF_2 to FVF_n, based on the plurality of trimming signals TRM_1 to TRM_{n-1}. Therefore, by applying the plurality of constant output voltages V_{O1} to V_{On} to the clock oscillate circuit **200**, the occurrence of clock skew between a plurality of clock signals CLK_1 to CLK_n generated by the clock generating circuit **10** may be reduced or prevented.

FIG. **5** is a circuit diagram illustrating another embodiment of an FVF buffer of an LDO regulator **100** according to an example embodiment.

Referring to FIG. **5**, another example of the circuit implementation of second to nth FVF buffers FVF_2 to FVF_n included in an output buffer circuit **130** included in the LDO regulator **100** according to an example embodiment may be seen. An example of circuit implementation illustrated in FIG. **5** may be similar to an example of circuit implementation illustrated in FIG. **4**, and thus, a difference therebetween will be mainly described.

In the example embodiment of FIG. **5**, unlike the embodiment of FIG. **4**, a k−1th trimming signal TRM_{k-1} may be applied to a sixth switching element P3 instead of a current source I_S. In some example embodiments, the sixth switching element P3 may include a plurality of target switching elements connected with one another in parallel.

Connections between the plurality of target switching elements may be controlled based on the k−1th trimming signal TRM_{k-1}. At this time, connections between the plurality of target switching elements may be controlled to adjust the number of target switching elements which operate based on the k−1th trimming signal TRM_{k-1}.

When the k−1th trimming signal TRM_{k-1} is a signal for control performed to increase a kth output voltage V_{Ok}, the number of target switching elements operating among the plurality of target switching elements may decrease. Accordingly, a width W_{P3} of the sixth switching element P3 may decrease. In some example embodiments, as in Equation 1, the kth output voltage V_{Ok} may be inversely proportional to the width W_{P3} of the sixth switching element P**3**, and thus, the kth output voltage V_{Ok} may increase.

When the k−1th trimming signal TRM_{k-1} is a signal for control performed to decrease the kth output voltage V_{Ok}, the number of target switching elements operating among the plurality of target switching elements may increase. Accordingly, the width W_{P3} of the sixth switching element P**3** may increase. In some example embodiments, as in Equation 1, the kth output voltage V_{Ok} may be inversely proportional to the width W_{P3} of the sixth switching element P**3**, and thus, the kth output voltage V_{Ok} may decrease.

When the k−1th trimming signal TRM_{k-1} is a signal for control performed to intactly maintain the kth output voltage V_{Ok}, the number of target switching elements operating among the plurality of target switching elements may be maintained identically or substantially identically. Accordingly, the kth output voltage V_{Ok} may be intactly maintained.

As described above, the LDO regulator **100** according to an example embodiment may generate a plurality of trimming signals TRM_1 to TRM_{n-1} generated by the trimming control circuit **120**, based on a plurality of output voltages V_{O1} to V_{On}, and may adjust the number of target switching elements operating among the plurality of target switching elements included in the second to nth FVF buffers FVF_2 to FVF_n of the output buffer circuit **130** to control constant levels of a plurality of output voltages V_{O1} to V_{On}, based on the plurality of trimming signals TRM_1 to TRM_{n-1}. Therefore, by applying the plurality of constant output voltages V_{O1} to V_{On} to the clock oscillate circuit **200**, the occurrence of clock skew between a plurality of clock signals CLK_1 to CLK_n generated by the clock generating circuit **10** may be prevented.

FIG. **6** is a timing diagram illustrating a voltage and a signal of a trimming control circuit of an LDO regulator **100** according to an example embodiment.

Referring to FIG. **6**, in the LDO regulator **100** according to an example embodiment, a timing diagram showing variations of first to fourth output voltages V_{O1} to V_{O4}, selection signals SEL**2** to SEL**4**, comparison signals (for example, first and second comparison signals) COM_1 and COM_2, and first to third trimming signals TRM_1 to TRM_3 may be seen.

Here, the selection signals SEL**2** to SEL**4** may include a second selection signal SEL**2** which allows a multiplexer **121** to output a second output voltage V_{O2}, a third selection signal SEL**3** which allows the multiplexer **121** to output a third output voltage V_{O3}, and a fourth selection signal SEL**4** which allows the multiplexer **121** to output a fourth output voltage V_{O4}.

At a first time T**1**, the second selection signal SEL**2** may be activated. At this time, the second output voltage V_{O2} may be greater than the first output voltage V_{O1} by a value of the offset voltage V_{OFS} or more, and thus, the first comparison signal COM_1 may have a first value. Accordingly, it may be seen that a value of the first trimming signal TRM_1 decreases, and thus, the second output voltage V_{O2} decreases. Also, the second FVF buffer FVF_2 may output the second output voltage V_{O2} which is reduced based on the first trimming signal TRM_1.

At a second time T**2**, the third selection signal SEL**3** may be activated. At this time, the third output voltage V_{O3} may be below the first output voltage V_{O1} by a value of the offset voltage V_{OFS} or more, and thus, the second comparison signal COM_2 may have a third value. Accordingly, it may be seen that a value of the second trimming signal TRM_2 increases, and thus, the third output voltage V_{O3} increases.

Also, the third FVF buffer FVF_3 may output the third output voltage V_{O3} which increases based on the second trimming signal TRM_2.

At a third time **T3**, the fourth selection signal **SEL4** may be activated. At this time, because the fourth output voltage V_{O4} is within the reference error range with respect to the first output voltage V_{O1}, the first comparison signal COM_1 may have a second value, and the second comparison signal COM_2 may have a fourth value. Accordingly, it may be seen that a value of the third trimming signal TRM_3 is maintained, and thus, the fourth output voltage V_{O4} is maintained. Also, the fourth FVF buffer FVF_4 may output the same fourth output voltage V_{O4}, based on the third trimming signal TRM_3.

At a fourth time **T4**, the second selection signal **SEL2** may be activated. At this time, the second output voltage V_{O2} may decrease at the first time **T1**, and thus, the second output voltage V_{O2} may be within the reference error range with respect to the first output voltage V_{O1}. Accordingly, the first comparison signal COM_1 may have the second value, and the second comparison signal COM_2 may have the fourth value. Accordingly, it may be seen that a value of the first trimming signal TRM_1 is maintained, and thus, the second output voltage V_{O2} is maintained. Also, the second FVF buffer FVF_2 may output the same second output voltage V_{O2}, based on the first trimming signal TRM_1.

At a fifth time **T5**, the third selection signal **SEL3** may be activated. At this time, even when the third output voltage V_{O3} may increase at the second time **T2**, the third output voltage V_{O3} may be below the first output voltage V_{O1} by a value of the offset voltage V_{OFS} or more, and thus, the second comparison signal COM_2 may have the third value. Accordingly, it may be seen that a value of the second trimming signal TRM_2 increases, and thus, the third output voltage VO_3 increases. Also, the third FVF buffer FVF_3 may output the third output voltage V_{O3} which increases based on the second trimming signal TRM_2.

At a sixth time **T6**, the fourth selection signal **SEL4** may be activated. Like the third time **T3**, the fourth output voltage V_{O4} may be within the reference error range with respect to the first output voltage V_{O1}, and thus, may be controlled to be equal or substantially equal to the third time **T3**, whereby the same or substantially the same fourth output voltage V_{O4} may be output.

At a seventh time **T7**, the second selection signal **SEL2** may be activated. Like the fourth time **T4**, the second output voltage V_{O1} may be within the reference error range with respect to the first output voltage V_{O1}, and thus, may be controlled to be equal or substantially equal to the fourth time **T4**, whereby the same or substantially the same second output voltage V_{O1} may be output.

At an eighth time **T8**, the third selection signal **SEL3** may be activated. At this time, the fifth output voltage V_{O3} may increase at the fifth time **T5**, and thus, the third output voltage V_{O3} may be within the reference error range with respect to the first output voltage V_{O1}. Accordingly, the first comparison signal COM_1 may have the second value, and the second comparison signal COM_2 may have the fourth value. Accordingly, it may be seen that a value of the second trimming signal TRM_2 is maintained, and thus, the third output voltage V_{O3} is maintained. Also, the third FVF buffer FVF_3 may output the same third output voltage V_{O3}, based on the second trimming signal TRM_2.

As described above, the LDO regulator **100** according to an example embodiment may sequentially output the second to nth output voltages V_{O1} to V_{On} by using the multiplexer **121**, sequentially generate the first to n−1th trimming signals TRM_1 to TRM_{n-1}, and sequentially adjust the second to nth output voltages V_{O1} to V_{On}. Accordingly, a voltage difference between the plurality of output voltages V_{O1} to V_{On} may be adjusted to within the reference error range, and thus, the occurrence of clock skew between the plurality of clock signals CLK_1 to CLK_n generated by the clock generating circuit **10** may be reduced or prevented.

FIG. **7** is a flowchart illustrating an operating method of an LDO regulator, according to an example embodiment.

Referring to FIG. **7**, in operation **S710**, the LDO regulator **100** may generate the error voltage E_O. The LDO regulator **100** may amplify a difference between the first output voltage V_{O1} and the reference voltage V_{REF} by using the amplifier **110** to generate the error voltage E_O.

In operation **S720**, the LDO regulator **100** may generate the first to n−1th trimming signals TRM_1 to TRM_{n-1}. The LDO regulator **100** may generate the first to n−1th trimming signals TRM_1 to TRM_{n-1} by using the trimming control circuit **120**, based on the plurality of output voltages V_{O1} to V_{On}. The trimming control circuit **120** may sequentially generate each of the first to n−1th trimming signals TRM_1 to TRM_{n-1}. A method of generating each of the first to n−1th trimming signals TRM_1 to TRM_{n-1} will be described below in more detail with reference to FIG. **8**.

FIG. **8** is a flowchart illustrating an operating method of a trimming control circuit of an LDO regulator, according to an example embodiment.

Referring to FIG. **8**, in operation **S810**, the trimming control circuit **120** may output the kth output voltage V_{Ok}, based on the selection signal SEL. The trimming control circuit **120** may output, through the multiplexer **121**, the kth output voltage V_{Ok} which is one of the second to nth output voltages V_{O1} to V_{On}, based on the selection signal SEL.

In operation **S820**, the trimming control circuit **120** may output the comparison signals COM_1 and COM_2. The trimming control circuit **120** may determine whether the kth output voltage V_{Ok} is within the reference error range with respect to the first output voltage V_{O1} and may output the comparison signals COM_1 and COM_2 representing a comparison result.

In operation **S830**, the trimming control circuit **120** may generate the k−1th trimming signal TRM_{k-1}, based on the selection signal SEL and the comparison signals COM_1 and COM_2. The trimming control circuit **120** may generate the k−1th trimming signal TRM_{k-1} corresponding to the selection signal SEL by using the trimming signal generator **123**, based on the comparison signals COM_1 and COM_2.

Returning to FIG. **7**, in step **S730**, the LDO regulator **100** may generate the first to nth output voltages V_{O1} to V_{On}. The LDO regulator **100** may generate the first to nth output voltages V_{O1} to V_{On} by using the output buffer circuit **130**. At this time, the output buffer circuit **130** may generate the first to nth output voltages V_{O1} to V_{On} by using the plurality of FVF buffers FVF_1 to FVF_n, based on the error voltage E_O and the first to n−1th trimming signals TRM_1 to TRM_{n-1}.

FIG. **9** is a block diagram illustrating a memory device **1** according to an example embodiment.

Referring to FIG. **9**, the memory device **1** according to an example embodiment may include a clock generating circuit **10** and a plurality of DRAM chips (for example, first to fourth DRAM chips) **21** to **24**.

The clock generating circuit **10** may operate like the clock generating circuit **10** described above with reference to FIGS. **1** to **8**.

A first clock signal CLK_1 generated by the clock generating circuit **10** may be applied to the first DRAM chip **21**. A second clock signal CLK_2 generated by the clock generating circuit **10** may be applied to the second DRAM chip **22**. A third clock signal CLK_3 generated by the clock generating circuit **10** may be applied to the third DRAM chip **23**. A fourth clock signal CLK_4 generated by the clock generating circuit **10** may be applied to the fourth DRAM chip **24**.

The plurality of DRAM chips **21** to **24** included in the memory device **1** may operate based on the plurality of clock signals CLK_1 to CLK_4 generated by the clock generating circuit described above with reference to FIGS. **1** to **8**, and thus, may reduce or prevent the occurrence of clock skew between the plurality of clock signals CLK_1 to CLK_4. Accordingly, a malfunction of the memory device **1** caused by clock skew may be reduced or prevented.

Hereinabove, some example embodiments have been described in the drawings and the specification. Example embodiments have been described by using the terms described herein, but this has been merely used for describing the inventive concepts and has not been used for limiting a meaning or limiting the scope of the inventive concepts. Therefore, it may be understood by those of ordinary skill in the art that various modifications and other example embodiments may be implemented from the inventive concepts.

One or more of the elements disclosed above may include or be implemented in one or more processing circuitries such as hardware including logic circuits; a hardware/software combination such as a processor executing software; or a combination thereof. For example, the processing circuitries more specifically may include, but is not limited to, a central processing unit (CPU), an arithmetic logic unit (ALU), a digital signal processor, a microcomputer, a field programmable gate array (FGPA), a System-on-Chip (SoC), a programmable logic unit, a microprocessor, application-specific integrated circuit (ASIC), etc.

While the inventive concepts have been particularly shown and described with reference to example embodiments thereof, it will be understood that various changes in form and details may be made therein without departing from the scope of the inventive concepts.

What is claimed is:

1. A low dropout (LDO) regulator configured to generate first to nth output voltages, where n is a natural number greater than or equal to 2, and each of the first to nth output voltages corresponds to a reference voltage, the LDO regulator comprising:

an amplifier configured to generate an error voltage based on the reference voltage and a first output voltage of the first to nth output voltages;

a trimming control circuit configured to generate first to (n−1)th trimming signals based on the first to nth output voltages; and

an output buffer circuit configured to generate the first to nth output voltages based on the error voltage and the first to (n−1)th trimming signals.

2. The LDO regulator of claim **1**, wherein the amplifier is configured to amplify a difference between the first output voltage of the first to nth output voltages and the reference voltage to generate the error voltage.

3. The LDO regulator of claim **1**, wherein the trimming control circuit comprises:

a multiplexer configured to output one of second to nth output voltages of the first to nth output voltages, based on a selection signal;

a comparison circuit configured to compare the first output voltage of the first to nth output voltages with a kth output voltage output by the multiplexer, to output a comparison signal, where k is a natural number greater than or equal to 2 and less than or equal to n; and

a trimming signal generator configured to generate a (k−1)th trimming signal of the first to (n−1)th trimming signals, based on the selection signal and the comparison signal.

4. The LDO regulator of claim **3**, wherein the trimming control circuit further comprises a low pass filter configured to remove noise from the first output voltage of the first to nth output voltages and supply a noise-removed first output voltage to the comparison circuit.

5. The LDO regulator of claim **3**, wherein

the multiplexer is configured to sequentially output the second to nth output voltages based on the selection signal, and

the trimming signal generator is further configured to sequentially generate the first to (n−1)th trimming signals based on the selection signal.

6. The LDO regulator of claim **3**, wherein the comparison circuit comprises:

a first power source configured to add an offset voltage to the first output voltage of the first to nth output voltages to output an upper voltage;

a second power source configured to subtract the offset voltage from the first output voltage of the first to nth output voltages to output a lower voltage;

a first comparator configured to compare the upper voltage with a kth output voltage of the first to nth output voltages, to generate a first comparison signal; and

a second comparator configured to compare the lower voltage with the kth output voltage output by the multiplexer to generate a second comparison signal.

7. The LDO regulator of claim **6**, wherein

the first comparator is configured to output the first comparison signal having a first value when the kth output voltage output by the multiplexer is greater than the upper voltage, and output the first comparison signal having a second value when the kth output voltage is less than or equal to the upper voltage, and

the second comparator is configured to output the second comparison signal having a third value when the kth output voltage output by the multiplexer is less than the lower voltage, and output the second comparison signal having a fourth value when the kth output voltage output by the multiplexer is greater than or equal to the lower voltage.

8. The LDO regulator of claim **7**, wherein the trimming signal generator is configured to,

generate the (k−1)th trimming signal of the first to (n−1)th trimming signals to decrease the kth output voltage when the first comparison signal has the first value,

generate the (k−1)th trimming signal of the first to (n−1)th trimming signals to increase the kth output voltage when the second comparison signal has the third value, and

generate the (k−1)th trimming signal of the first to (n−1)th trimming signals to maintain the kth output voltage when the first comparison signal has the second value and the second comparison signal has the fourth value.

9. The LDO regulator of claim **1**, wherein the output buffer circuit is configured to,

control a bias current based on the first to (n−1)th trimming signals, and

generate the first to nth output voltages based on the bias current.

10. The LDO regulator of claim 1, wherein the output buffer circuit is configured to,

control connections between a plurality of target switching elements based on the first to (n−1)th trimming signals, and

generate the first to nth output voltages based on voltages of the plurality of target switching elements.

11. The LDO regulator of claim 1, wherein the output buffer circuit comprises:

a first flipped voltage follower (FVF) buffer configured to generate a first output voltage based on the error voltage; and

second to nth FVF buffers configured to generate second to nth output voltages based on the error voltage and the first to (n−1)th trimming signals.

12. The LDO regulator of claim 11, wherein each of the second to nth FVF buffers comprises a current source configured to adjust a bias current, based on each of the first to (n−1)th trimming signals.

13. The LDO regulator of claim 11, wherein

each of the second to nth FVF buffers comprises a plurality of target switching elements connected with one another in parallel, and

connections between the plurality of target switching elements are controlled based on the first to (n−1)th trimming signals.

14. A clock generating circuit comprising:

a low dropout (LDO) regulator configured to generate a plurality of output voltages each corresponding to a reference voltage; and

a clock oscillate circuit configured to generate a plurality of clock signals based on the plurality of output voltages,

wherein the LDO regulator comprises,

an amplifier configured to generate an error voltage based on the reference voltage and a first output voltage of the plurality of output voltages,

a trimming control circuit configured to generate a plurality of trimming signals based on the plurality of output voltages, and

an output buffer circuit configured to generate the plurality of output voltages based on the error voltage and the plurality of trimming signals.

15. The clock generating circuit of claim 14, wherein the trimming control circuit comprises:

a multiplexer configured to output one of second to nth output voltages of the plurality of output voltages based on a selection signal;

a comparison circuit configured to compare the first output voltage of the first to nth output voltages with a kth output voltage output by the multiplexer, to output a comparison signal, where k is a natural number greater than or equal to 2 and less than or equal to n; and

a trimming signal generator configured to generate a (k−1)th trimming signal of the plurality of trimming signals, based on the selection signal and the comparison signal.

16. The clock generating circuit of claim 15, wherein

the multiplexer is configured to sequentially output the second to nth output voltages based on the selection signal, and

the trimming signal generator is configured to sequentially generate the plurality of trimming signals based on the selection signal.

17. The clock generating circuit of claim 15, wherein the comparison circuit comprises:

a first power source configured to add an offset voltage to the first output voltage of the first to nth output voltages, to output an upper voltage;

a second power source configured to subtract the offset voltage from the first output voltage of the first to nth output voltages, to output a lower voltage;

a first comparator configured to compare the upper voltage with a kth output voltage of the plurality of output voltages, to generate a first comparison signal; and

a second comparator configured to compare the lower voltage with the kth output voltage output by the multiplexer to generate a second comparison signal.

18. The clock generating circuit of claim 14, wherein the output buffer circuit is configured to,

control a bias current based on the plurality of trimming signals, and

generate the plurality of output voltages based on the bias current.

19. The clock generating circuit of claim 14, wherein the output buffer circuit is configured to,

control connections between a plurality of target switching elements based on the plurality of trimming signals, and

generate the plurality of output voltages based on voltages of the plurality of target switching elements.

20. A memory device comprising:

a clock generating circuit configured to generate a plurality of clock signals; and

a plurality of dynamic random access memory (DRAM) chips configured to operate based on the plurality of clock signals,

wherein the clock generating circuit comprises,

a low dropout (LDO) regulator configured to generate a plurality of output voltages each corresponding to a reference voltage, and

a clock oscillate circuit configured to generate the plurality of clock signals, based on the plurality of output voltages, and

the LDO regulator comprises,

an amplifier configured to generate an error voltage based on the reference voltage and a first output voltage of the plurality of output voltages,

a trimming control circuit configured to generate a plurality of trimming signals based on the plurality of output voltages, and

an output buffer circuit configured to generate the plurality of output voltages based on the error voltage and the plurality of trimming signals.

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